Huawei Access Node WFD-ACC-09 QoS Congestion Event Resolution Procedure

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7830

Description

The QoS congestion event failure domain on the Huawei Access Node WFD-ACC-09 at the Whitefield POP is primarily attributed to the occurrence of microbursts, interface tail drops, and QoS queue starvation. This phenomenon can be explained by the interplay of multiple factors, including inadequate queuing configurations, insufficient buffer sizes, and suboptimal queue scheduling strategies. In the context of Huawei's NE40E platform, the access node's role as a critical juncture for multiple service flows necessitates the deployment of sophisticated QoS mechanisms to ensure optimal resource allocation and efficient traffic handling. However, if the QoS configuration is not properly optimized, it can lead to congestion events, characterized by microbursts, interface tail drops, and QoS queue starvation. To mitigate this issue, it is essential to employ a systematic approach to troubleshooting, leveraging Huawei's advanced diagnostic tools and CLI commands to identify the root cause of the congestion event and implement targeted corrective actions.

Pre-requisite Checks

- Ensure the Whitefield POP is accessible via a reliable communication link.

- Verify that the redundant link is operational and can assume traffic responsibility in case of a failure.

- Dispatch remote hands to the Whitefield POP to facilitate on-site troubleshooting and maintenance activities.

Potential Root Causes

Cause	Probability
Inadequate queuing configurations	High
Insufficient buffer sizes	Medium
Suboptimal queue scheduling strategies	Low

Diagnostic Commands & Expected Outputs

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Command: display queue statistics, Expected Output: Queue statistics for each interface and queue will be displayed., Description: This command will provide detailed information about the current queuing configurations, including buffer sizes, queue scheduling strategies, and packet loss rates. By analyzing the output, it is possible to identify potential bottlenecks and areas for optimization.

Command: display qos statistics, Expected Output: QoS statistics for each service flow will be displayed., Description: This command will provide detailed information about the current QoS configurations,
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including queue scheduling strategies, packet loss rates, and service flow packet rates. By analyzing the output, it is possible to identify potential QoS issues and areas for optimization.

Command: display interface counters, Expected Output: Interface counters for each interface will be displayed., Description: This command will provide detailed information about the current interface counters, including packet loss rates, packet discard rates, and error rates. By analyzing the output, it is possible to identify potential interface-related issues and areas for optimization.
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Resolution Steps

Step 1: Step: Optimize queuing configurations, Command: queue-set optimize, Description: This command will optimize the queuing configurations for each interface, ensuring that buffer sizes are sufficient and queue scheduling strategies are optimal. By optimizing the queuing configurations, it is possible to prevent congestion events and ensure efficient traffic handling.

Step 2: Step: Adjust buffer sizes, Command: queue-set buffer-size adjust, Description: This command will adjust the buffer sizes for each interface, ensuring that they are sufficient to handle the current traffic loads. By adjusting the buffer sizes, it is possible to prevent packet loss and ensure efficient traffic handling.

Step 3: Step: Modify queue scheduling strategies, Command: queue-set scheduling-strategy modify, Description: This command will modify the queue scheduling strategies for each interface, ensuring that they are optimal for the current traffic patterns. By modifying the queue scheduling strategies, it is possible to prevent QoS issues and ensure efficient traffic handling.

Post-Check Verification

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Command: display queue statistics, Expected Output: Queue statistics for each interface and queue will be displayed with optimized values., Description: This command will verify that the queuing configurations have been optimized and that buffer sizes are sufficient. By analyzing the output, it is possible to confirm that the congestion event has been mitigated.

Command: display qos statistics, Expected Output: QoS statistics for each service flow will be displayed with optimized values., Description: This command will verify that the QoS configurations have been optimized and that service flows are being handled efficiently. By analyzing the output, it is possible to confirm that the congestion event has been mitigated.

Command: display interface counters, Expected Output: Interface counters for each interface will be displayed with optimized values., Description: This command will verify that the interface counters have been optimized and that packet loss rates, packet discard rates, and error rates are within acceptable limits. By analyzing the output, it is possible to confirm that the congestion event has been mitigated.
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Roll-back Procedures

- Step: Revert queuing configurations, Command: queue-set revert, Description: This command will revert the queuing configurations to their original values, ensuring that the system is returned to its previous state. By reverting the queuing configurations, it is possible to restore the system to its original operational state.

- Step: Reset buffer sizes, Command: queue-set buffer-size reset, Description: This command will reset the buffer sizes to their original values, ensuring that the system is returned to its previous state. By resetting the buffer sizes, it is possible to restore the system to its original operational state.

- Step: Restore queue scheduling strategies, Command: queue-set scheduling-strategy restore, Description: This command will restore the queue scheduling strategies to their original values, ensuring that the system is returned to its previous state. By restoring the queue scheduling strategies, it is possible to restore the system to its original operational state.